

Topic in Module 1 — Edge Computing

Course	Modern Computing and Python Programming (26ESCC106)
Module	Module 1 — Modern Computing Approaches
Topic coverage	Edge Computing — Introduction, Need, Architecture, Cloud vs. Edge, Applications
Position in module	After Cloud Computing; last topic of the module
Mapped CO	CO1 — Describe modern computing approaches and their characteristics
Bloom's level	L1 — Remember, L2 — Understand
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1. Introduction

1.1 Definition

Edge computing means performing computation close to where the data is produced, instead of sending all of it to a distant data centre [4, 1]. The “edge” is the outer boundary of the network: the place where cameras, sensors, machines, vehicles and phones meet the network, as opposed to the centre, where the large cloud data centres sit.

Two terms are worth fixing before going further, because the whole topic turns on them.

- **Latency** is the time taken for data to travel from where it is produced to where it is processed, and for the answer to come back. It is measured in milliseconds (ms), and one millisecond is a thousandth of a second.
- **Bandwidth** is how much data a network link can carry per second, measured in megabits per second (Mbps). A link with higher bandwidth carries more data in the same time.

In the cloud model studied in the previous topic, data travels from the device to a data centre that may be hundreds or thousands of kilometres away, is processed there, and the result travels back. Edge computing places a computer — a gateway, a small server, or the device itself — beside the data source, so that most of the work finishes locally and only a small part of the data ever travels to the cloud. Figure 1.1 shows this basic arrangement.

Edge computing does not replace cloud computing. The two work together: the edge handles what must be fast, private or independent of the network, and the cloud handles what needs large storage, heavy computation or a view across many sites.

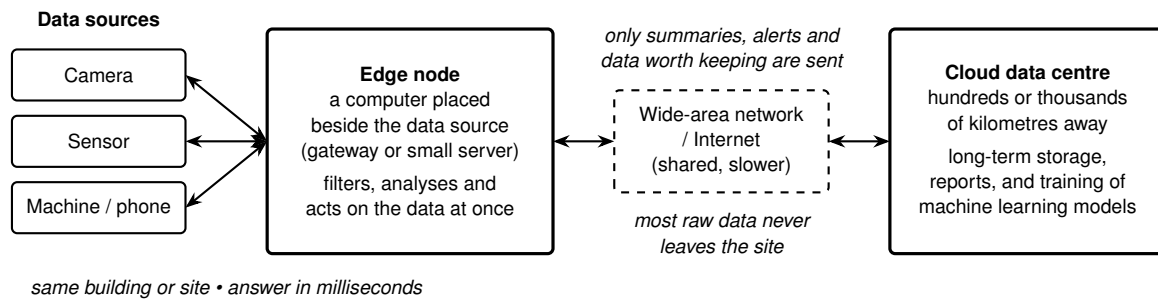


Figure 1.1: Basic working model of edge computing

1.2 Where the Edge Is

“The edge” is not one fixed place. Depending on the application the computing may sit at any of the following points, and a real system often uses more than one of them [1].

- **On the device itself** — a camera that recognises a number plate inside the camera, or a watch that counts steps without help from a phone
- **On a gateway at the site** — one small computer in a factory, farm or hospital that serves all the devices around it
- **At the network operator’s premises** — a small data centre at a mobile base station, serving every user connected to it. This arrangement is called multi-access edge computing [7]

The term **fog computing** is often met alongside edge computing. It describes the same idea with the emphasis on an intermediate layer of gateways and regional nodes between the devices and the cloud, rather than on the devices themselves [6]. For this course the two may be treated as the same approach seen from slightly different distances.

2. Need for Edge Computing

Cloud computing solved the problem of obtaining capacity without owning it. It did not solve the problem of distance. Sending data to a distant data centre takes time, consumes network capacity, and fails when the network fails. The reasons for computing at the edge follow from these facts [4, 5].

- **Latency** — some decisions cannot wait for a round trip to a data centre. A self-driving vehicle that must brake, a robot arm that must stop before it injures someone, or a game that must respond to a movement, needs an answer within a few milliseconds. A round trip to a cloud region typically costs tens of milliseconds, and more when the network is busy.
- **Bandwidth** — sending everything is often simply too much data. One high-definition camera produces roughly 4 Mbps of video, so a campus with 100 cameras would need about 400 Mbps of upload capacity, continuously, only to send video that is almost entirely uneventful. An edge node can watch all 100 streams locally and send on only the few seconds that matter.
- **Intermittent connectivity** — ships, mines, farms, oil rigs and rural clinics do not have a dependable link to the Internet. Work that depends on reaching the cloud stops when the link does; work done at

the edge continues and synchronises later.

- **Privacy and regulation** — medical records, classroom footage and factory process data may be sensitive, or may be required by law to stay within the premises or the country. Processing locally and sending only anonymous summaries keeps the raw data where it belongs.
- **Cost** — network transfer and cloud storage are charged by volume, so discarding uninteresting data at the edge reduces both bills.
- **Scale** — the number of connected devices grows faster than the capacity of the links to the cloud, so a design that forwards everything to the centre becomes harder to sustain as devices are added [1, 3].

3. Edge Computing Architecture

An edge computing system is usually described as a set of tiers, or layers, arranged by distance from the data [1, 6]. Figure 1.2 shows the four tiers and what each one does. It is read from the bottom, where the data is created, to the top, where the cloud sits.

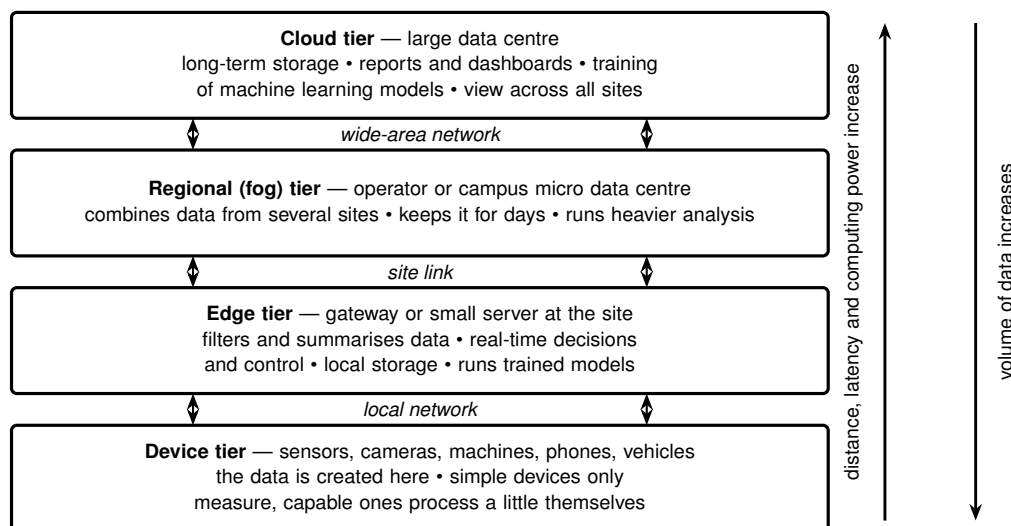


Figure 1.2: Architecture of an edge computing system: four tiers, from the devices that create the data to the cloud

3.1 The Tiers

- **Device tier** — the sensors, cameras, machines, vehicles and phones that produce the data. A simple sensor only measures and reports; a more capable device performs a little processing itself, such as discarding a reading that has not changed.
- **Edge tier** — one or more computers at the site: a gateway, an industrial PC or a small server. This tier receives data from the devices over a local network, filters and summarises it, takes the decisions that must be immediate, keeps a short history locally, and forwards what is worth forwarding.
- **Regional or fog tier** — a micro data centre belonging to the organisation or to the network operator and serving several sites. It combines their data, keeps it for days or weeks, and runs analysis that is too heavy for a gateway. Small systems often leave this tier out and connect the edge tier directly to the cloud.

- **Cloud tier** — the large data centre studied in the previous topic. It provides effectively unlimited storage and computing power, a view across every site, and the place where machine learning models are trained before being sent down to the edge to be used.

3.2 How Work Is Divided Between the Tiers

Two quantities change as one moves up the diagram, and the design follows from them. The volume of data is largest at the bottom, because every device produces data continuously, and it shrinks at each tier as data is filtered and summarised. Computing power, storage and distance all grow towards the top. The working rule is therefore simple: perform a task at the lowest tier that is able to perform it, and send upward only what the higher tier genuinely needs [4].

A traffic camera illustrates the division. The camera captures video (device tier). The edge node watches that video, recognises a vehicle crossing a red light and raises an alert within a fraction of a second, discarding the rest (edge tier). Each evening it uploads the recorded violations and a count of vehicles per hour to the cloud, which stores them, prepares reports for the whole city, and periodically trains an improved recognition model that is then sent back to the edge nodes (cloud tier).

4. Cloud versus Edge Computing

The two are partners rather than rivals, but they differ in almost every practical respect [4, 2].

Aspect	Cloud computing	Edge computing
Location of processing	Large data centres, far from the user	Close to where the data is produced
Typical response time	Tens of milliseconds or more	A few milliseconds
Computing power	Very large and elastic	Limited by what fits at the site
Storage	Effectively unlimited	Small; usually a short local history
Data sent over the network	All of it	Only summaries and important events
Works without the Internet	No	Yes, for local tasks
Privacy of raw data	Leaves the premises	Can stay on the premises
Cost pattern	Pay for capacity, transfer and storage used	Pay once for the equipment, then little transfer cost
Scope of view	Every site together	One site or one device
Suitable work	Training models, long-term storage, reports across sites	Immediate decisions, filtering, local control
Management	One provider manages the hardware	Many small units in many places must be managed

5. Applications

- **Industrial automation** — machines on a production line are monitored for vibration, temperature and current, and an edge node stops a machine or raises a maintenance alert before a fault becomes a breakdown
- **Autonomous and assisted driving** — a vehicle interprets its cameras and sensors on board, because braking cannot wait for a reply from a data centre [5]
- **Video surveillance and traffic management** — cameras and edge nodes detect events locally and send only the relevant clips instead of streaming everything
- **Healthcare** — bedside monitors raise alarms locally, and patient data is summarised before anything leaves the hospital
- **Agriculture** — soil, weather and crop-health sensors are processed on a farm gateway, which keeps working where network coverage is poor
- **Smart cities and utilities** — street lighting, water distribution and electricity meters are controlled locally, with readings summarised for the central system
- **Retail** — shelf cameras, billing counters and stock sensors are processed in the store, so that it continues to operate even if the link to the head office is lost

- **Content delivery and gaming** — copies of popular videos, pages and game servers are kept near users so that they load quickly and the central servers carry less load

6. Advantages and Limitations

Advantages

- Responses in milliseconds, because the data does not travel far
- Much less network traffic, and lower transfer and storage cost
- Continues to work when the connection to the cloud is lost
- Sensitive data can be kept within the premises
- Capacity grows naturally as devices and sites are added

Limitations

- Computing power, memory and storage at a site are limited compared with a data centre
- Many small units spread over many locations are harder to update, monitor and repair than one data centre
- Physical security is weaker: equipment sits in factories, streets and vehicles rather than in a guarded building
- Devices and gateways from different makers must be made to work together, and standards are still settling
- Only a local view is available, so conclusions that need data from every site still require the cloud

7. Review Questions

CO1 — Describe modern computing approaches and their characteristics

Bloom's Level L1 — Remember, L2 — Understand

1. Define edge computing. Where exactly is “the edge” in a computing system?
2. Define latency and bandwidth, and state the unit in which each is measured.
3. With a neat diagram, explain the basic working model of edge computing.
4. State any four reasons why edge computing is needed.
5. Explain, with one example each, how edge computing helps when the network connection is unreliable and when the data is sensitive.
6. With a neat diagram, explain the architecture of an edge computing system, naming each tier and stating what it does.
7. What is a gateway in an edge computing system, and what work does it perform?

8. Differentiate between cloud computing and edge computing with respect to location of processing, response time, storage and network usage.
9. Explain why edge computing is described as a partner of cloud computing rather than a replacement for it.
10. What is fog computing, and how is it related to edge computing?
11. List any five applications of edge computing, stating in each case why the processing must be done locally.
12. State any four limitations of edge computing.
13. Describe how a traffic camera system would divide its work between the device, edge and cloud tiers.

8. References and Further Reading

Items 1 to 3 are the prescribed books for this module; chapter numbers are indicative. Items 4 to 6 are the papers from which the definitions and the tiered architecture are taken, and item 7 is the standards site for edge computing in mobile networks.

- [1] P. Lea, *IoT and Edge Computing for Architects*, 2nd Edition, Packt Publishing, 2020 — Reference Book 1 for this course; chapters on edge computing and edge architecture.
- [2] R. Buyya, C. Vecchiola, S. T. Selvi, S. Poojara and S. N. Srirama, *Mastering Cloud Computing*, 2nd Edition, Morgan Kaufmann, 2026 — Text Book 2 for this course; used here for the cloud side of the comparison.
- [3] K. Hwang, G. C. Fox and J. J. Dongarra, *Distributed and Cloud Computing: From Parallel Processing to the Internet of Things*, Morgan Kaufmann, 2012 — Text Book 1 for this course; used here for sensors, devices and the growth in the number of connected things.
- [4] W. Shi, J. Cao, Q. Zhang, Y. Li and L. Xu, “Edge Computing: Vision and Challenges”, *IEEE Internet of Things Journal*, vol. 3, no. 5, pp. 637–646, 2016.
- [5] M. Satyanarayanan, “The Emergence of Edge Computing”, *IEEE Computer*, vol. 50, no. 1, pp. 30–39, 2017.
- [6] F. Bonomi, R. Milito, J. Zhu and S. Addepalli, “Fog Computing and Its Role in the Internet of Things”, *Proceedings of the First MCC Workshop on Mobile Cloud Computing*, ACM, 2012, pp. 13–16.
- [7] ETSI, Multi-access Edge Computing (MEC). <https://www.etsi.org/technologies/multi-access-edge-computing>